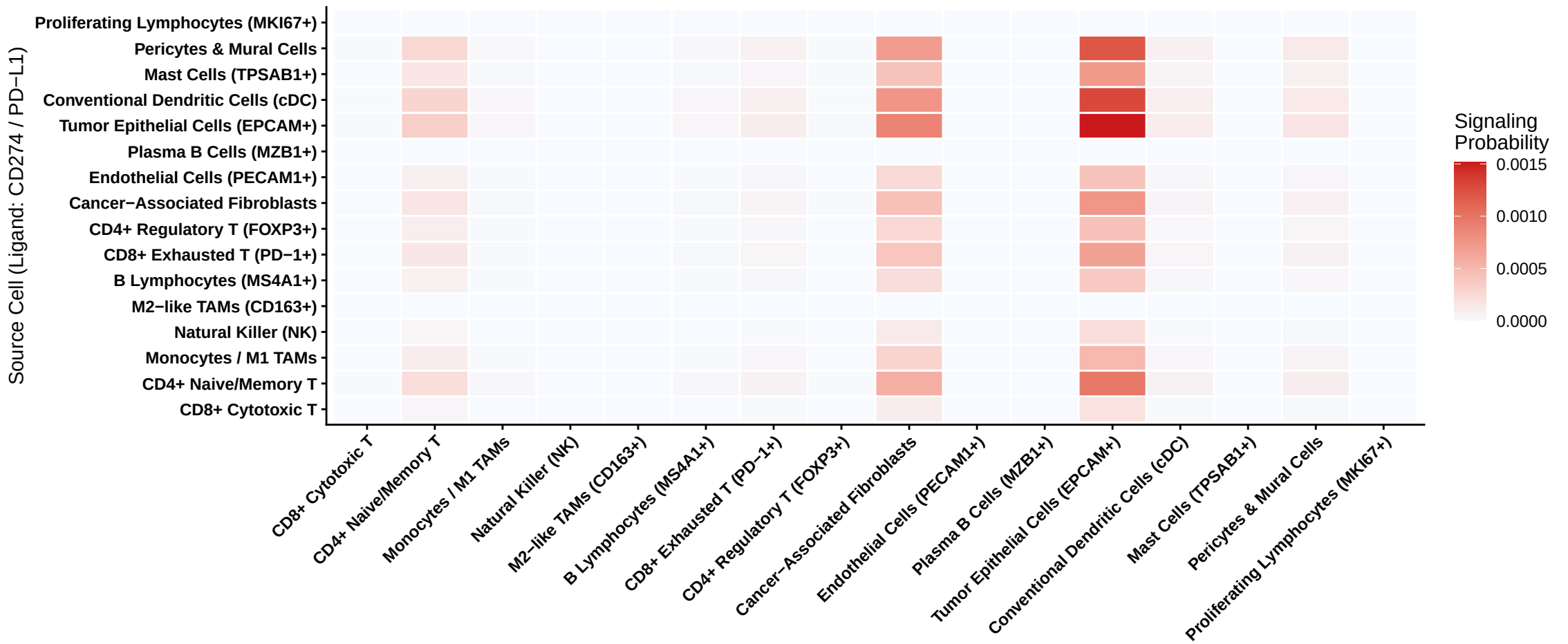
Figure 4: Intercellular Signaling Interactome & Mass−Action Network Centrality

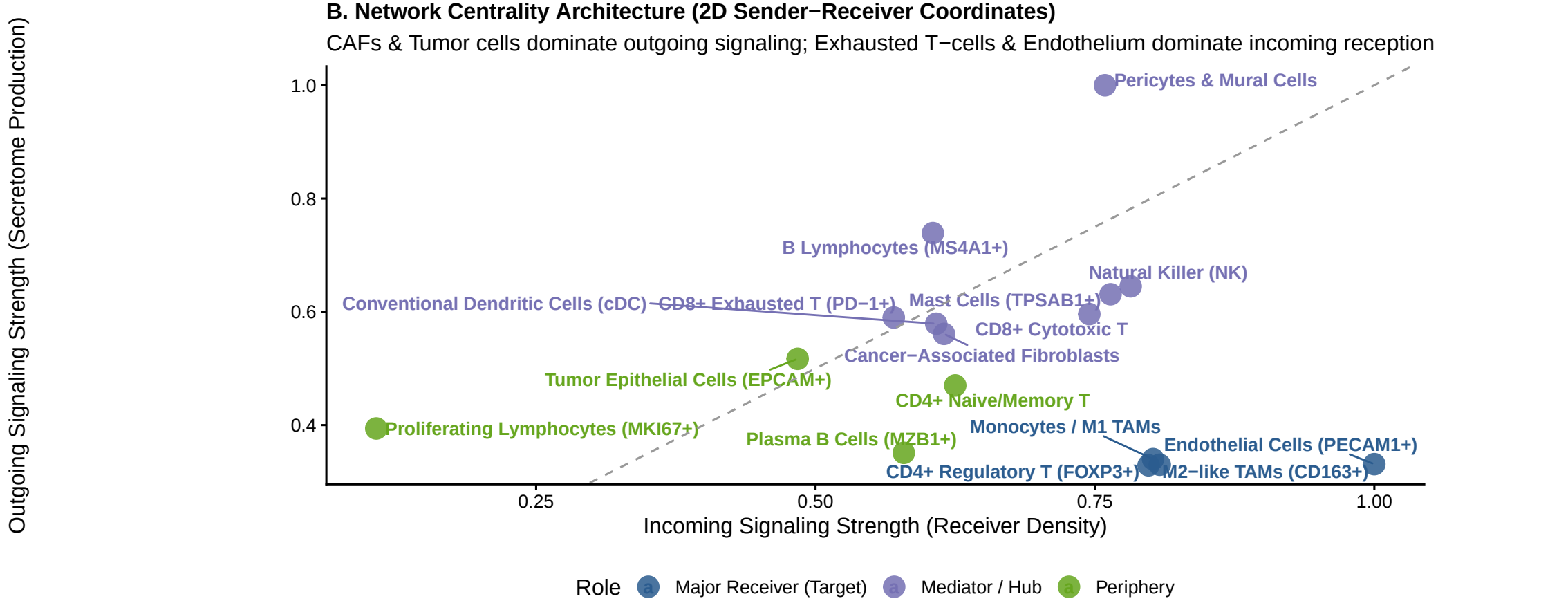
Modeled via CellChat v2 framework across 12 tumor microenvironment cell populations

A. PD−L1/PD−1 Checkpoint Axis

Targeting CD8+ Exhausted TILs by M2 TAMs & Epithelial Tumor Cells



Target Cell (Receptor: PDCD1)

B. Network Centrality Architecture (2D Sender−Receiver Coordinates)

CAFs & Tumor cells dominate outgoing signaling; Exhausted T−cells & Endothelium dominate incoming reception